RaptorVision: Few-Shot Object Detection via Dense Patch Embeddings and Cosine Similarity in DINOv3 Feature Space

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Abstract

RaptorVision is a training-free, few-shot object detection system designed for high-throughput analysis of wildlife camera trap imagery. Rather than fine-tuning a detector on labelled examples, it leverages the rich, semantically structured feature space produced by DINOv3 vision transformers. A user selects one or a handful of representative image patches; RaptorVision then computes the cosine similarity between the embedding of each query patch and every spatial position in subsequent images, producing a dense similarity map that highlights regions containing objects of the same visual class. This paper explains the internal mechanics of the system: the feature extraction pipeline, the geometry of the embedding space, the similarity computation, and end with some discussions on how we may improve the algorithm.

Contents

1 Introduction **2**

2 Background **2**
2.1 Vision Transformers and Patch Tokenisation . . . 2
2.2 DINOv3 Self-Supervised Pretraining . . . 3
2.3 The Unit Hypersphere and Cosine Similarity . . . 3

3 Feature Extraction Pipeline **4**
3.1 Image Preprocessing . . . 4
3.2 Dense Feature Map Extraction . . . 5

4 Query and Similarity Computation **5**
4.1 Patch Selection via User Click . . . 5
4.2 Single-Image Cosine Similarity Map . . . 6
4.3 Library-Based Similarity (Memory View) . . . 6

5 Empirical Evaluation **7**
5.1 Experimental Setup . . . 7
5.2 Per-Dataset Results . . . 8
5.2.1 Biome Health Project — Maasai Mara 2018 . . . 8
5.2.2 California Small Animals . . . 10
5.2.3 Ohio Small Animals (OSU) . . . 12
5.2.4 Snapshot Serengeti . . . 14
5.3 Quantitative Summary . . . 16

5.4 Cross-Dataset Generalisation . . . 16
5.5 Limitations . . . 18

6 Toward Richer Class Representations: Beyond Cosine Similarity 18
6.1 The Implicit Probabilistic Model of Cosine Similarity . . . 18
6.2 Mahalanobis Distance in the Ambient Embedding Space . . . 19
6.3 Fisher-Bingham Distribution on the Hypersphere . . . 20
6.4 Geometric Interpretation: Preferred and Penalised Directions . . . 20
6.5 Practical Considerations . . . 21

7 Conclusion 21

1 Introduction

Monitoring wildlife with camera traps generates large volumes of images, the vast majority of which contain no animal of interest. Manually reviewing thousands of frames is impractical, and training a conventional object detector requires labelled bounding boxes—a scarce resource in field ecology. RaptorVision addresses this bottleneck through a fundamentally different paradigm: *few-shot, training-free detection*.

The key insight is that modern self-supervised vision transformers, and DINOv3 in particular, produce feature representations with a powerful semantic structure: patches depicting visually or semantically similar content are mapped to nearby directions in the embedding space. Detection therefore reduces to a nearest-neighbour search: given one or a few example patches chosen by the user, any image location whose embedding is sufficiently close to a query embedding is flagged as a match.

No gradient computation, no labelled dataset, and no retraining are required. The pretrained DINOv3 weights remain entirely frozen; only the similarity threshold is tunable by the user.

2 Background

2.1 Vision Transformers and Patch Tokenisation

A Vision Transformer (ViT) [1] processes an image by first partitioning it into a regular grid of non-overlapping *patches*. In DINOv3, each patch covers a 16×16 pixel region. For an image of dimensions $H \times W$ (both multiples of 16), this yields a sequence of $h_p \times w_p$ tokens, where $h_p = H/16$ and $w_p = W/16$.

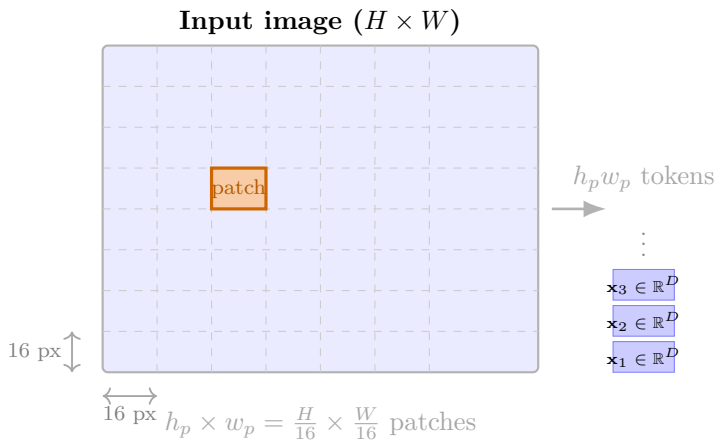


Figure 1: An image of dimensions $H \times W$ is partitioned into $h_p \times w_p$ non-overlapping 16×16 pixel patches. Each patch is linearly projected into a D-dimensional token. The orange cell illustrates a single selected patch.

Each patch is linearly projected to a D-dimensional vector, and the full sequence is processed by a stack of multi-head self-attention layers. Crucially, the output at each spatial position retains information about the entire image context, not only the local patch.

2.2 DINOv3 Self-Supervised Pretraining

DINOv3 [2] is trained with a self-distillation objective on the LVD-1689M dataset—a curated collection of approximately 1.7 billion images spanning a wide variety of visual domains.

The consequence of this pretraining regime is a strong *semantic clustering* property: patches that depict similar visual concepts (fur texture, bark, water surface, ...) are mapped to nearby directions in the embedding space, regardless of the specific image from which they originate. RaptorVision exploits this property directly.

Three model sizes are available, differing in embedding dimensionality D:

Variant	**Architecture**	**Dimension D**
Small	ViT-S/16	384
Base	ViT-B/16	768
Large	ViT-L/16	1024

Table 1: DINOv3 model variants supported by RaptorVision.

2.3 The Unit Hypersphere and Cosine Similarity

After extraction, every feature vector is L2-normalised, projecting it onto the unit hypersphere $\mathcal{S}^{D-1} \subset \mathbb{R}^D$:

$$\hat{\mathbf{u}} = \frac{\mathbf{u}}{\|\mathbf{u}\|_2}, \qquad \|\hat{\mathbf{u}}\|_2 = 1.$$

The cosine similarity between two vectors $\mathbf{u}$ and $\mathbf{v}$ is defined as:

$$\cos(\mathbf{u}, \mathbf{v}) = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\|_2 \, \|\mathbf{v}\|_2}.$$

Because all vectors are unit-normalised, this simplifies to a plain dot product:

$$\cos(\hat{\mathbf{u}}, \hat{\mathbf{v}}) = \hat{\mathbf{u}} \cdot \hat{\mathbf{v}} = \cos\theta,$$

where θ is the angle between the two vectors. The similarity ranges from -1 (opposite directions) to $+1$ (identical direction), and is 0 for orthogonal vectors.

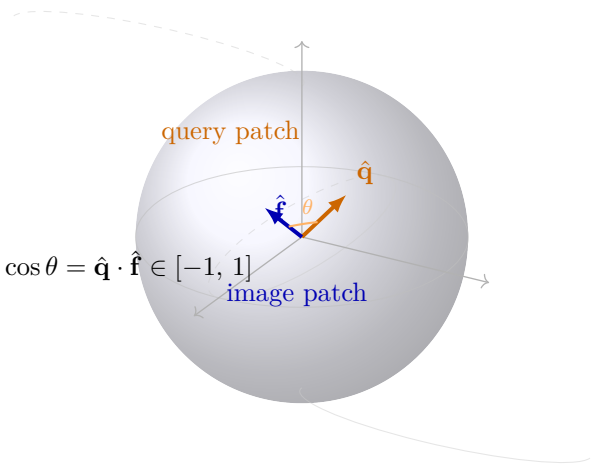


Figure 2: All L2-normalised embeddings lie on the unit hypersphere $\mathcal{S}^{D-1}$. The cosine similarity between the query vector $\hat{\mathbf{q}}$ (orange) and an image patch vector $\hat{\mathbf{f}}$ (blue) equals the cosine of the angle θ between them, which in turn equals their dot product.

This geometric interpretation is central to RaptorVision: detection is equivalent to finding all positions on the sphere that are within a certain angular distance of the query vector.

3 Feature Extraction Pipeline

3.1 Image Preprocessing

Before feature extraction, each image undergoes a deterministic preprocessing sequence designed to satisfy the ViT's structural requirements while limiting memory consumption.

Aspect-preserving resize. The image is scaled so that its largest dimension does not exceed a configurable maximum size (default: 672 pixels), while preserving the original aspect ratio:

$$r = \min\left(\frac{S_{\max}}{W}, \frac{S_{\max}}{H}\right), \qquad W' = rW, \quad H' = rH.$$

16-pixel alignment. Both dimensions are then rounded down to the nearest multiple of 16 to ensure exact alignment with the ViT patch grid:

$$\tilde{W} = 16\left\lfloor\frac{W'}{16}\right\rfloor, \qquad \tilde{H} = 16\left\lfloor\frac{H'}{16}\right\rfloor.$$

Normalisation. The pixel values are normalised using the ImageNet channel statistics to match the distribution seen during DINOv3 pretraining:

$$\tilde{x}_c = \frac{x_c - \mu_c}{\sigma_c}, \qquad \boldsymbol{\mu} = (0.485, 0.456, 0.406), \quad \boldsymbol{\sigma} = (0.229, 0.224, 0.225).$$

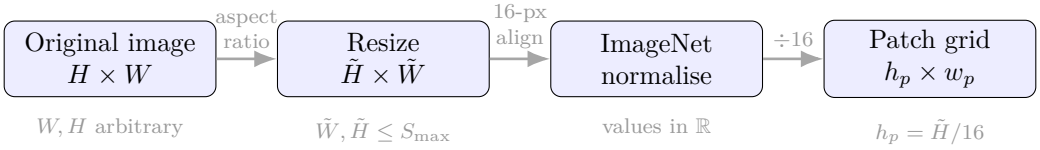


Figure 3: The preprocessing pipeline applied to every image before DINOv3 inference. The two constraints—maximum size and 16-pixel alignment—ensure exact correspondence between pixel coordinates and patch-grid indices.

3.2 Dense Feature Map Extraction

The preprocessed image tensor is passed through the DINOv3 backbone. Instead of using the final classification token (`[CLS]`), RaptorVision extracts the output of the last intermediate transformer layer, which retains the full spatial token sequence. The output has shape $(1,\, h_p w_p,\, D)$.

After discarding the `[CLS]` token and retaining only the $h_p w_p$ spatial tokens, the sequence is reshaped into a 2D spatial grid and L2-normalised along the feature dimension:

$$\mathbf{F} \;\in\; \mathbb{R}^{h_p \times w_p \times D}, \qquad \|\mathbf{F}[i,j,:]\|_2 = 1 \;\; \forall\, i,j.$$

This produces a *dense feature map*: a 2D array in which each cell contains a unit-normalised D-dimensional embedding representing the visual content of the corresponding 16×16 pixel region.

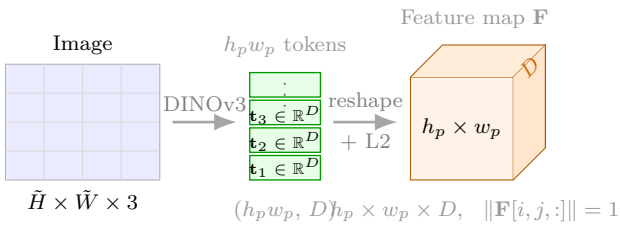


Figure 4: The DINOv3 backbone converts a preprocessed image into a dense feature map $\mathbf{F} \in \mathbb{R}^{h_p \times w_p \times D}$. Each cell of the grid holds a unit-normalised D-dimensional embedding for the corresponding 16×16 pixel region.

4 Query and Similarity Computation

4.1 Patch Selection via User Click

When the user clicks at pixel coordinates (x, y) in the displayed image, RaptorVision maps this position to the corresponding cell in the feature grid. The screen coordinates are first converted to normalised image coordinates:

$$r_x = \frac{x}{W_{\text{display}}}, \qquad r_y = \frac{y}{H_{\text{display}}},$$

and then to integer grid indices:

$$p_x = \min(\lfloor r_x \cdot w_p \rfloor,\; w_p - 1)\,, \qquad p_y = \min(\lfloor r_y \cdot h_p \rfloor,\; h_p - 1)\,.$$

The query vector is the feature at that grid cell:

$$\mathbf{q} = \mathbf{F}[p_y,\, p_x,\, :] \;\in\; \mathbb{R}^D, \qquad \|\mathbf{q}\|_2 = 1.$$

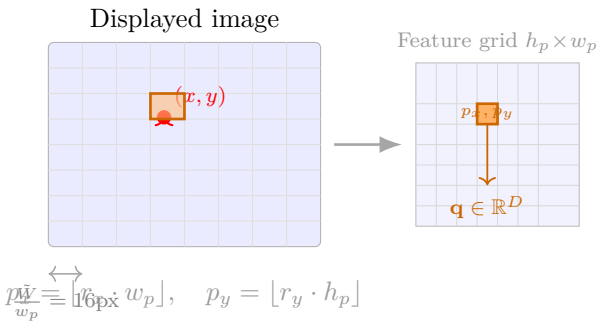


Figure 5: A click at pixel (x, y) is converted to normalised image coordinates (r_x, r_y) and then mapped to grid indices (p_x, p_y). The corresponding feature vector $\mathbf{q}$ is extracted as the query embedding.

4.2 Single-Image Cosine Similarity Map

Once the query vector $\mathbf{q}$ is available, the cosine similarity between $\mathbf{q}$ and every spatial position of the current image is computed as a single matrix-vector product. The feature map $\mathbf{F}$ is first flattened along the spatial dimensions:

$$\mathbf{F}_{\text{flat}} \in \mathbb{R}^{(h_p w_p) \times D}.$$

The similarity vector is then:

$$\mathbf{s} = \mathbf{F}_{\text{flat}}\, \mathbf{q} \;\in\; \mathbb{R}^{h_p w_p},$$

and is reshaped back to (h_p, w_p) to form a 2D *similarity heatmap*. Because all vectors are unit-normalised, each entry $s[i, j] = \cos(\mathbf{F}[i, j, :], \mathbf{q})$ directly measures the angular proximity between the image patch and the query.

4.3 Library-Based Similarity (Memory View)

When the user has built a *patch library* containing K reference patches—each represented by its unit-normalised embedding—the system computes similarity against all library vectors simultaneously. Let $\mathbf{L} \in \mathbb{R}^{K \times D}$ denote the stacked library matrix. The joint similarity computation is:

$$\mathbf{S} = \mathbf{F}_{\text{flat}}\, \mathbf{L}^{\top} \;\in\; \mathbb{R}^{(h_p w_p) \times K}.$$

Each row $\mathbf{S}[n, :]$ contains the similarity scores between image patch n and all K library patches. The best-matching library patch for each image position is found by a max-reduction along the library dimension:

$$s^*[n] = \max_{k=1,\ldots,K} \mathbf{S}[n, k], \qquad n = 1, \ldots, h_p w_p.$$

The result $\mathbf{s}^* \in \mathbb{R}^{h_p w_p}$ is reshaped to (h_p, w_p) and rendered as a heatmap.

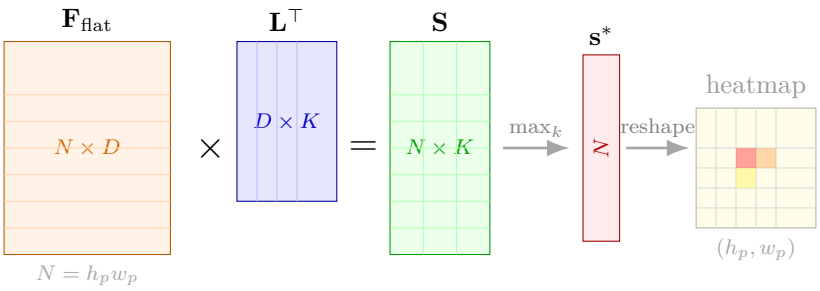


Figure 6: Library similarity computation. The flattened feature map $\mathbf{F}_{\text{flat}}$ $(N \times D)$ is multiplied by the transposed library matrix $\mathbf{L}^\top$ $(D \times K)$, yielding a joint similarity matrix $\mathbf{S}$ $(N \times K)$. A max-reduction over the library dimension gives the best library match per image patch, which is reshaped into the 2D heatmap.

5 Empirical Evaluation

5.1 Experimental Setup

All experiments use the **DINOv3-Base** backbone (ViT-B/16, $D = 768$) with an input resolution of **720 pixels**. The DINOv3-Large backbone at 1,024 pixels is expected to yield higher performance but was excluded in order to minimise computational overhead during the evaluation phase.

Protocol. Each dataset is split into a **100-image training set** and a **900-image test set**, sampled from the animal-present class. This highly asymmetric 10/90 split is intentional: it replicates the realistic constraint of scarce labelled data and directly exercises RaptorVision's few-shot capability. From the 100 training images the system constructs a patch memory; the effect of memory size N (number of stored patches) on detection quality is studied by varying N. A **control group** of approximately 500 blank (empty) images is included in every evaluation to measure the false-positive rate.

Metrics. Rather than reporting performance at a single operating point, we present the full Receiver Operating Characteristic (ROC) curve and its Area Under the Curve (AUC) as the primary quantitative metric. The optimal decision threshold is identified via Youden's index ($J = \text{TPR} - \text{FPR}$) to jointly minimise miss-rate and false-alarm rate.

Datasets. Table 2 summarises the four public camera-trap datasets used in this evaluation. They span African savanna ecosystems (Maasai Mara, Serengeti), North American woodland and grassland small-mammal deployments (California), and a challenging herpetofauna/infrared dataset (Ohio/OSU), providing broad coverage of ecological contexts and imaging modalities.

Table 2: Summary of evaluation datasets. All datasets are sourced from lila.science. "Train" images are used exclusively for patch memory construction; "Test" and "Control" images are never seen during memory building.

Dataset	Cat.	Train	Test	Control
Biome Health Project — Maasai Mara 2018	53	116	823	630
California Small Animals	74	90	607	1,000
Ohio Small Animals (OSU)	–	90	803	610
Snapshot Serengeti	33	96	824	1,000

5.2 Per-Dataset Results

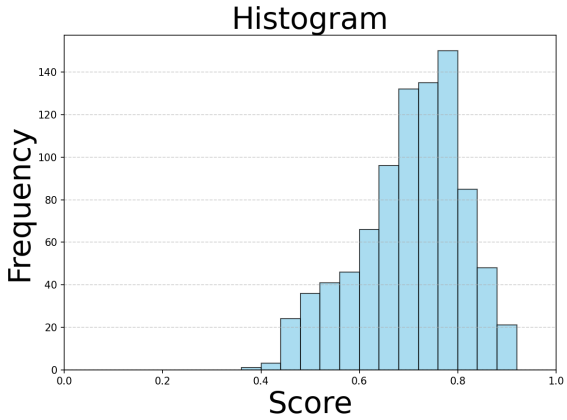
5.2.1 Biome Health Project — Maasai Mara 2018

This dataset spans 53 distinct species categories across the Maasai Mara ecosystem. Representative samples are shown in Figure 7.

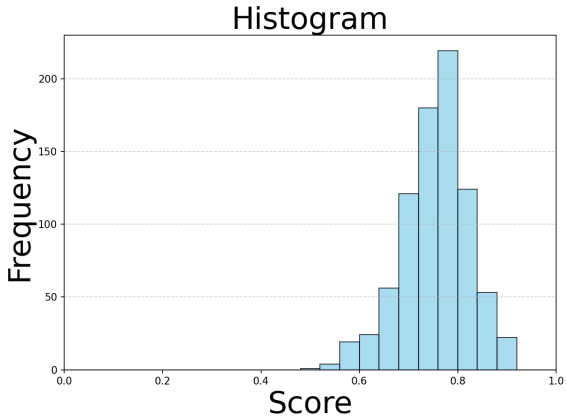
Figure 7: Representative images from the Biome Health Project — Maasai Mara 2018 dataset. The collection spans large ungulates, predators, and small mammals in a savanna context.

Figure 8 shows the score distributions as a function of patch memory size N. A clear bimodal separation emerges as N increases: animal-present images cluster near high scores while empty images accumulate at lower values. The corresponding ROC curve (Figure 9), evaluated at $N = 44$ patches, yields an **AUC of 0.9618** with an optimal threshold of **0.66** (TPR $= 0.91$, FPR $= 0.08$).

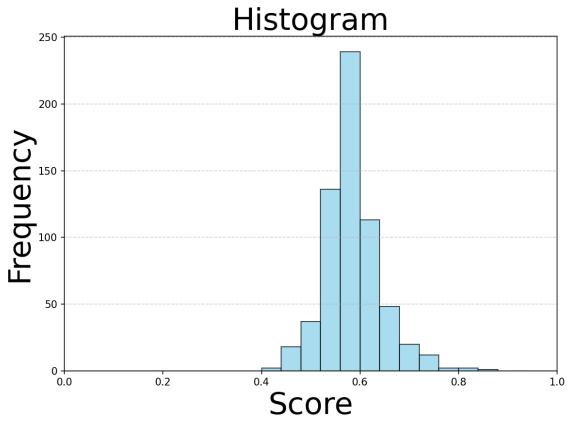
Score outliers in the lower tail of the animal class are largely attributable to annotation noise in the source dataset. Typical causes include images where the labelled animal is absent, partially occluded, or too small to be resolved at 720-pixel resolution (Figure 10).



(a) 10 patches



(b) 44 patches (animal class)



(c) 44 patches (empty/control class)

Figure 8: Score distributions for the Maasai Mara dataset at two memory sizes. Class separation improves monotonically with N; the control class (empty images) remains tightly concentrated at low scores throughout.

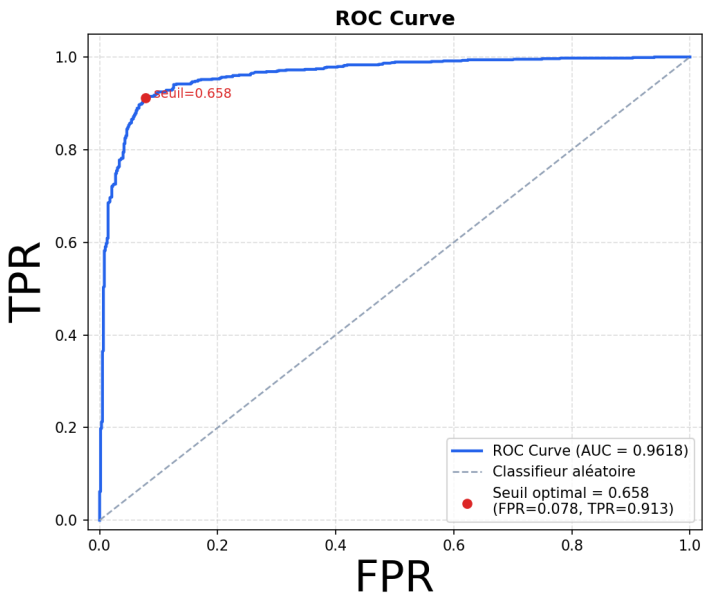


Figure 9: ROC curve for the Maasai Mara dataset ($N = 44$ patches). AUC $= 0.9618$; optimal threshold $= 0.66$.

Score: 0.501

Score: 0.522

Score: 0.550

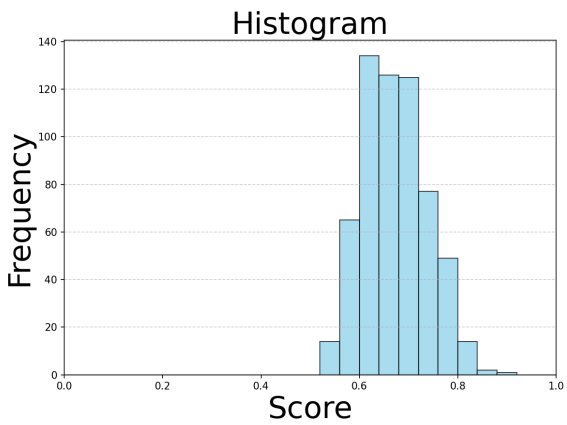
Figure 10: Three lowest-scoring animal-class images from the Maasai Mara test set ($N = 44$). Low scores are consistent with annotation noise, severe occlusion, or subjects too distant to resolve at 720 px.

5.2.2 California Small Animals

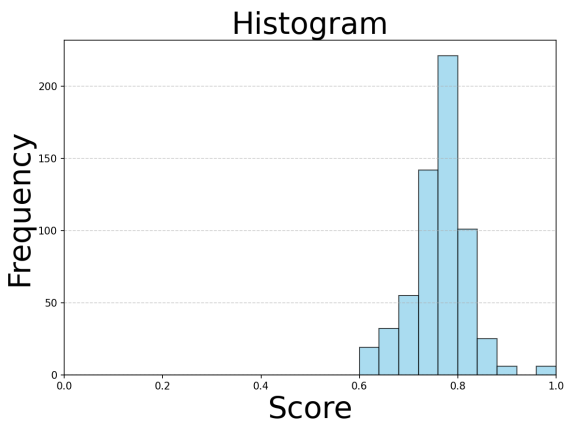
This dataset covers 74 species categories of small mammals and birds in California. The high category count and small average subject size make it a challenging benchmark (Figure 11).

Figure 11: Representative images from the California Small Animals dataset.

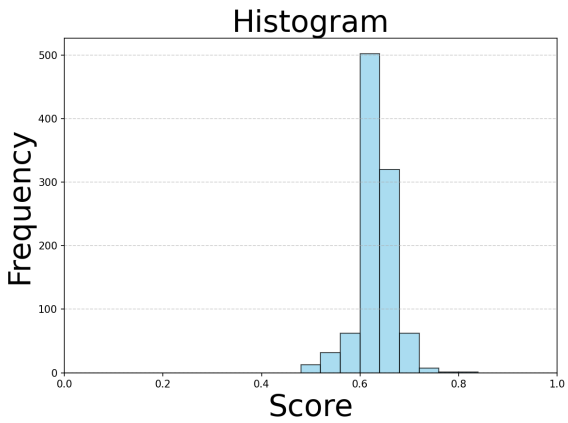
Despite the high species diversity and small subject size, RaptorVision achieves an **AUC of 0.9689** with an optimal threshold of **0.68** (TPR = 0.92, FPR = 0.06) using a 40-patch memory (Figures 12 and 13). The slightly higher optimal threshold compared to other datasets is consistent with a tendency for small subjects to produce moderately elevated scores on empty frames that share similar background texture.



(a) 5 patches



(b) 40 patches (animal class)



(c) 40 patches (empty/control class)

Figure 12: Score distributions for the California Small Animals dataset. Even with 74 categories, the patch memory achieves robust class separation.

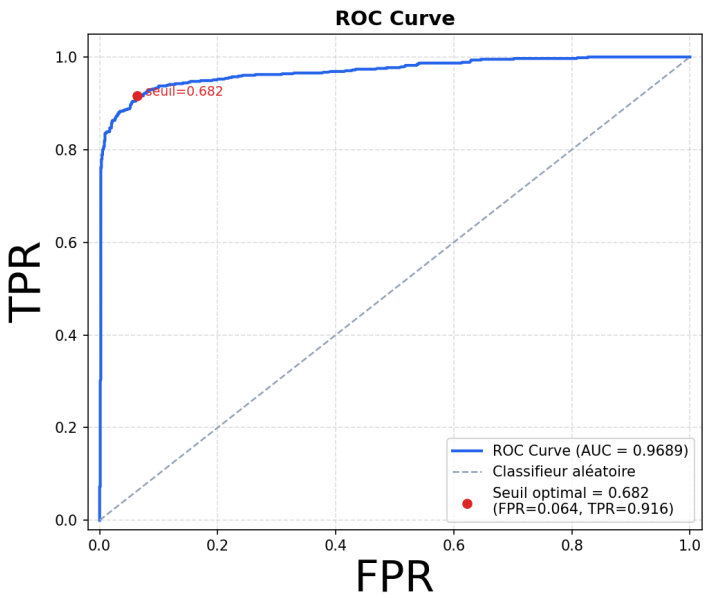


Figure 13: ROC curve for the California Small Animals dataset ($N = 40$ patches). AUC $= 0.9689$; optimal threshold $= 0.68$.

Score: 0.609

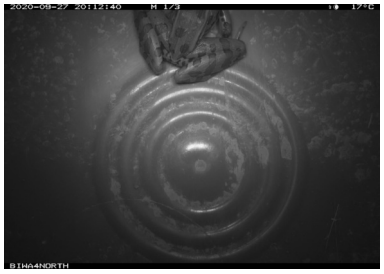
Score: 0.611

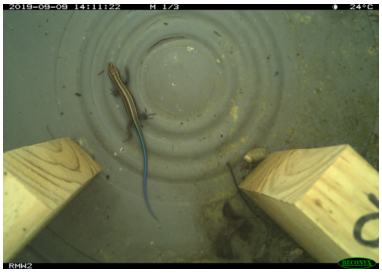
Score: 0.620

Figure 14: Three lowest-scoring animal-class images from the California Small Animals test set ($N = 40$). Small subject size and cluttered backgrounds are the primary causes of reduced scores.

5.2.3 Ohio Small Animals (OSU)

The OSU dataset focuses on herpetofauna—primarily frogs, toads, and lizards—in nocturnal and low-visibility near-infrared conditions (Figure 15). This constitutes one of the most challenging settings in the evaluation: subjects are small, camouflaged, and frequently appear at low contrast against leaf-litter backgrounds under infrared illumination.





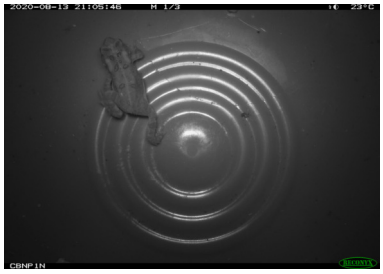
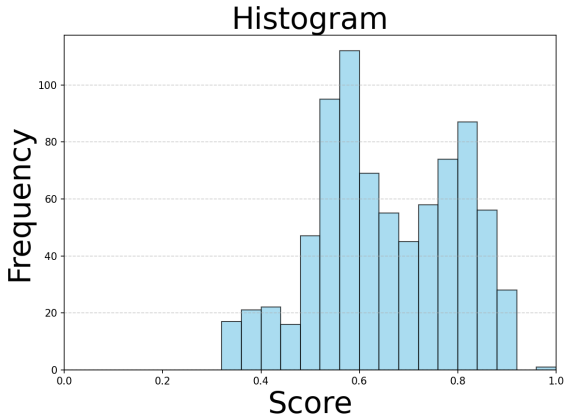
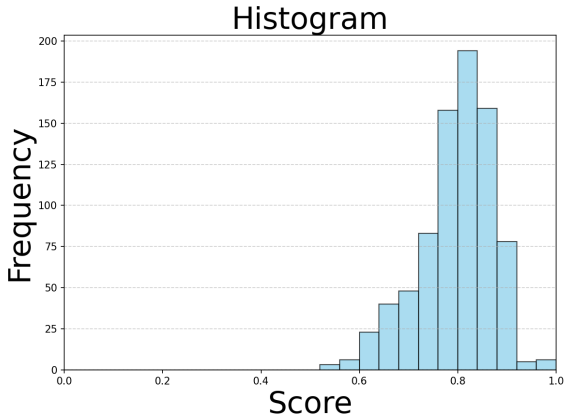


Figure 15: Representative images from the Ohio Small Animals (OSU) dataset. Note the near-infrared imaging modality and low-contrast subjects against leaf-litter backgrounds.

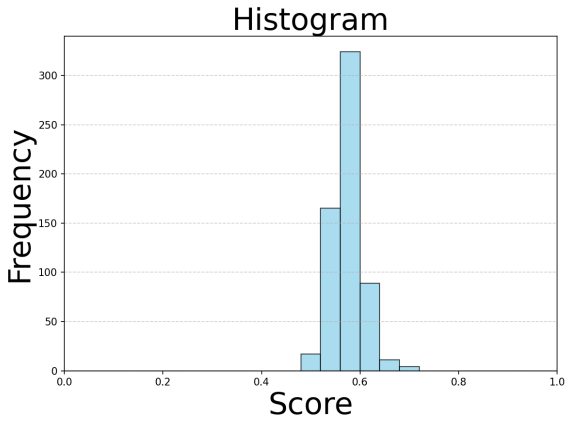
Despite these challenges, RaptorVision achieves an **AUC of 0.9918** with an optimal threshold of **0.65** (TPR = 0.95, FPR = 0.01)—the second-highest AUC in the benchmark—using only **9 patches** (Figures 16 and 17). This result is particularly noteworthy: DINOv3 was pretrained on RGB images, yet its representations generalise effectively to near-infrared imagery without any domain adaptation.



(a) 2 patches



(b) 9 patches (animal class)



(c) 9 patches (empty/control class)

Figure 16: Score distributions for the OSU dataset. Strong class separation is achieved with as few as 9 patches, indicating that the infrared-modality herpetofauna class is highly compact in DINOv3 feature space.

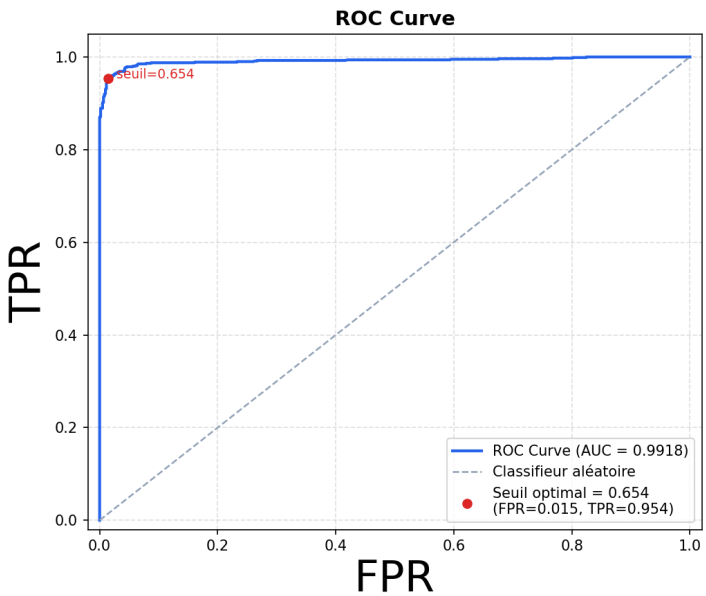


Figure 17: ROC curve for the OSU dataset ($N = 9$ patches). AUC = 0.9918; optimal threshold = 0.65.

Score: 0.541 Score: 0.545 Score: 0.551

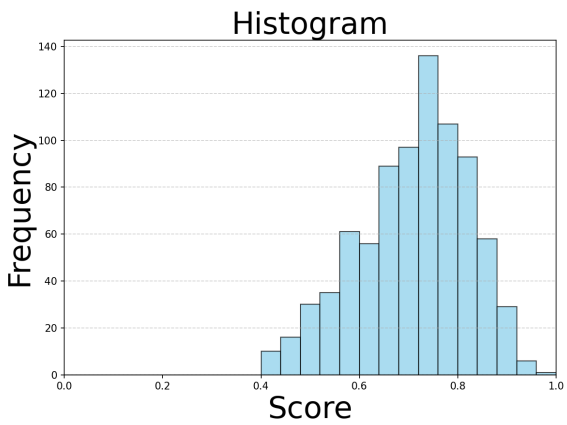
Figure 18: Three lowest-scoring animal-class images from the OSU test set ($N = 9$). Extreme camouflage, very small subject size, and flash-induced overexposure are the principal failure modes.

5.2.4 Snapshot Serengeti

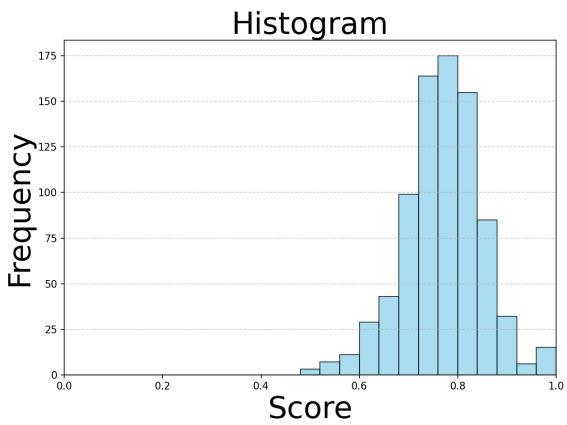
The Snapshot Serengeti dataset encompasses 33 species of large African mammals. Its wide diversity of species sizes, behaviours, and backgrounds provides a comprehensive test of in-ecosystem generalisation (Figure 19).

Figure 19: Representative images from the Snapshot Serengeti dataset.

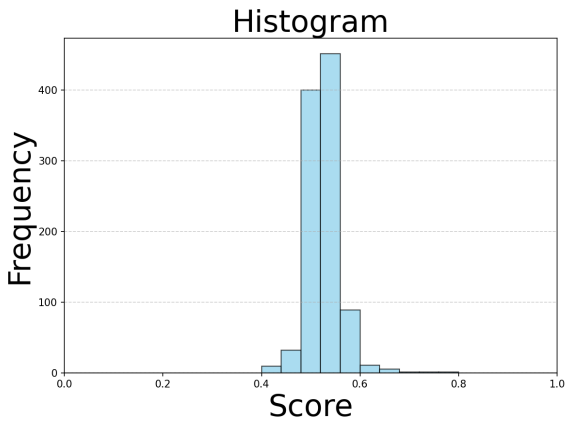
RaptorVision achieves its highest single-dataset AUC on this benchmark: **AUC = 0.9932** with an optimal threshold of **0.60** (TPR = 0.97, FPR = 0.02) using a 42-patch memory (Figures 20 and 21). The lower optimal threshold compared to other datasets likely reflects the larger apparent size of megafauna subjects, which produce stronger and more consistent patch activations even at 720-pixel resolution.



(a) 5 patches



(b) 42 patches (animal class)



(c) 42 patches (empty/control class)

Figure 20: Score distributions for the Snapshot Serengeti dataset. Near-perfect class separation is achieved with a 42-patch library.

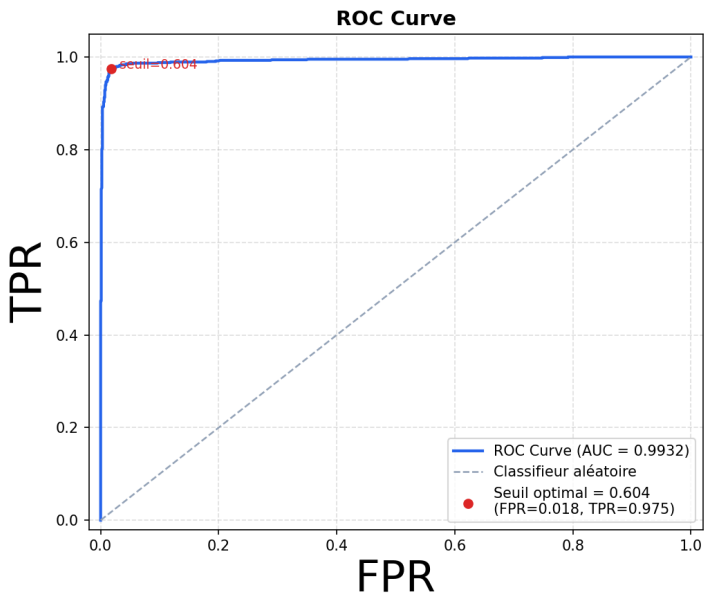


Figure 21: ROC curve for the Snapshot Serengeti dataset ($N = 42$ patches). AUC = 0.9932; optimal threshold = 0.60.

Score: 0.505 Score: 0.508 Score: 0.516

Figure 22: Three lowest-scoring animal-class images from the Snapshot Serengeti test set ($N = 42$). Animals that are distant, heavily backlit, or occupying only a small fraction of the frame yield the lowest scores.

5.3 Quantitative Summary

Table 3 consolidates the quantitative results across all four datasets. RaptorVision consistently achieves AUC values above 0.96, regardless of ecosystem type, species diversity, or imaging modality. Two datasets exceed 0.99, demonstrating near-perfect discrimination with only 100 training images.

A key operational finding is that the optimal decision threshold remains stable in the [0.60, 0.70] range across all benchmarks. This stability means that a single threshold can serve as a practical default without per-deployment calibration.

Table 3: Quantitative results across all benchmark datasets at maximum patch count N. All experiments use DINOv3-Base at 720 px input resolution and a 100-image training set.

Dataset	N	**AUC**	**Threshold**	**TPR**	**FPR**
Biome Health Project — Maasai Mara 2018	44	0.9618	0.66	0.91	0.08
California Small Animals	40	0.9689	0.68	0.92	0.06
Ohio Small Animals (OSU)	9	0.9918	0.65	0.95	0.01
Snapshot Serengeti	42	0.9932	0.60	0.97	0.02

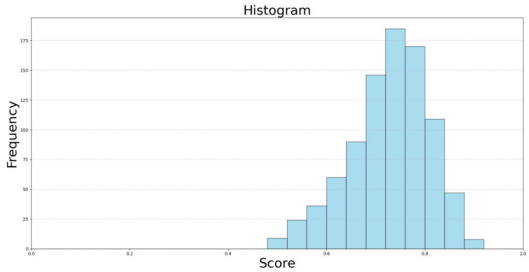
5.4 Cross-Dataset Generalisation

A critical practical question is whether a patch library built in one ecological context retains discriminative power when applied to a different deployment. To assess this, we conducted two reciprocal transfer experiments between the Snapshot Serengeti and the Maasai Mara datasets—two geographically distinct but ecologically comparable African savanna ecosystems.

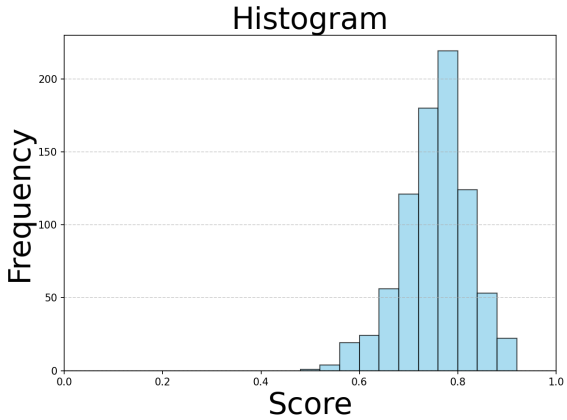
Serengeti library on Maasai Mara. The 42-patch library built from Snapshot Serengeti images was applied directly to the Maasai Mara test set without modification. Figure 23 shows that the resulting score distribution remains well separated, demonstrating that DINOv3 patch representations learned in one savanna biome transfer to another without any retraining.

Maasai Mara library on Serengeti. The reciprocal experiment applied the 44-patch Maasai Mara library to the Serengeti test set. As shown in Figure 24, the cross-domain score distribution remains competitive with the in-domain reference, further confirming the robustness of the patch-matching approach.

These results confirm that RaptorVision's feature representations generalise across ecological contexts. In practice, a library built from one deployment can serve as a reliable starting point

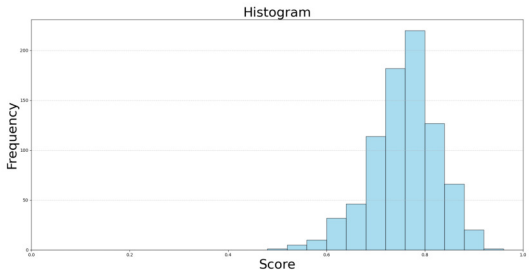


(a) Serengeti library → Maasai Mara test set (cross-dataset)

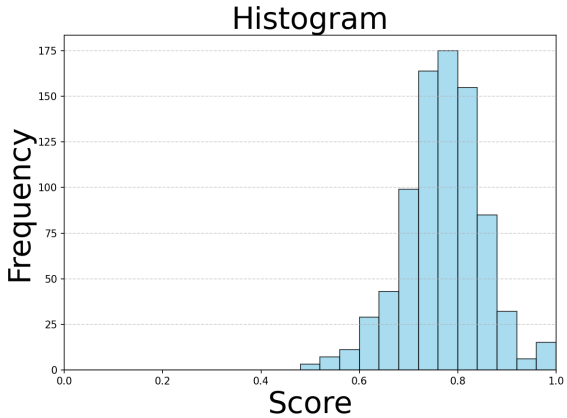


(b) Maasai Mara library → Maasai Mara test set (in-domain reference)

Figure 23: Cross-dataset transfer: Snapshot Serengeti patch library evaluated on the Maasai Mara dataset, compared against the in-domain baseline. Class separation is well preserved despite the geographic and species-composition differences.



(a) Maasai Mara library → Serengeti test set (cross-dataset)



(b) Serengeti library → Serengeti test set (in-domain reference)

Figure 24: Cross-dataset transfer: Maasai Mara patch library evaluated on the Snapshot Serengeti dataset, compared against the in-domain baseline.

for a new deployment in a comparable environment, further reducing the annotation burden in the field.

5.5 Limitations

Taxonomic granularity. The evaluation focuses on the binary task of distinguishing animal-present from empty images. When tested on the finer-grained task of differentiating morphologically similar species within the same taxonomic family (specifically, distinct frog species in the OSU dataset), RaptorVision currently cannot perform reliable fine-grained species classification. When two species share similar texture and morphology, the DINOv3 features that discriminate "animal" from "background" do not necessarily capture the subtle inter-species differences. Addressing this limitation would require either species-specific patch libraries with tighter labelling, or a secondary classification head fine-tuned on species-level annotations.

Input resolution and subject size. Reduced scores for small or distant subjects (see Figures 10, 14, 18 and 22) indicate that the 720-pixel input resolution may be insufficient when animals occupy fewer than approximately 5% of the image area. Upgrading to DINOv3-Large at 1,024-pixel resolution is expected to mitigate this issue at the cost of increased inference time.

6 Toward Richer Class Representations: Beyond Cosine Similarity

The detection criterion used in RaptorVision—thresholding the cosine similarity—is conceptually simple and computationally cheap, but it rests on a strong geometric assumption: that a visual class occupies a *symmetric, isotropic* region of the embedding sphere. This section analyses why this assumption is an approximation, and how two principled statistical tools—the Mahalanobis distance and the Fisher-Bingham distribution—can overcome it by modelling the true anisotropic shape of a class in DINOv3 feature space.

6.1 The Implicit Probabilistic Model of Cosine Similarity

Thresholding the dot product $\hat{\mathbf{f}} \cdot \hat{\mathbf{q}} > \tau$ is the hard decision boundary of the **von Mises-Fisher (vMF)** distribution, the natural analogue of the isotropic Gaussian on the unit hypersphere $\mathcal{S}^{D-1}$:

$$p(\mathbf{x}\,;\,\boldsymbol{\mu},\kappa) \;\propto\; \exp\!\big(\kappa\,\boldsymbol{\mu}^{\top}\mathbf{x}\big), \qquad \mathbf{x} \in \mathcal{S}^{D-1},$$

where $\boldsymbol{\mu}$ is the mean direction and $\kappa \geq 0$ is the concentration parameter. The density depends *only* on the dot product with $\boldsymbol{\mu}$, making it perfectly isotropic: every direction at the same angular distance from $\boldsymbol{\mu}$ receives identical probability mass. Fixing a threshold $\tau = \cos\theta_0$ delimits a *spherical cap*—the set of all unit vectors within angle θ_0 of $\boldsymbol{\mu}$.

When K library vectors are combined via the max-reduction of Section 4.3, the detection region becomes the *union* of K spherical caps, one per library patch. Each cap remains individually isotropic, so the overall model is still a collection of symmetric, round decision regions.

(a) Cosine similarity — vMF (isotropic)

(b) Fisher-Bingham — anisotropic

Figure 25: Both panels show the same 3D unit hypersphere under identical perspective. (a) Cosine similarity thresholding defines a *circular* cap: all tangential directions around $\boldsymbol{\mu}$ are penalised equally. (b) A real visual class forms an *elongated elliptical* region on the sphere. The major axis $\boldsymbol{\gamma}_2$ (red, long arrow) spans the direction of within-class variation (lighting, scale, viewpoint); the class is wide here. The minor axis $\boldsymbol{\gamma}_3$ (purple, short arrow) points toward the boundary with neighbouring classes; the class is tight here. Both axes lie in the tangent space at $\boldsymbol{\gamma}_1$. The Fisher-Bingham term $\beta[(\boldsymbol{\gamma}_2^\top \mathbf{x})^2 - (\boldsymbol{\gamma}_3^\top \mathbf{x})^2]$ captures this asymmetry; cosine similarity (vMF) cannot.

The isotropic cap of cosine similarity therefore leads to two systematic failure modes:

- **False negatives** along $\boldsymbol{\gamma}_2$: genuine patches of the class that lie further than θ_0 from $\boldsymbol{\gamma}_1$ due to legitimate within-class variation are rejected.
- **False positives** along $\boldsymbol{\gamma}_3$: patches of a different class that are geometrically close to $\boldsymbol{\gamma}_1$ but displaced in the tight discriminative direction are incorrectly accepted.

6.2 Mahalanobis Distance in the Ambient Embedding Space

A first improvement is to model the class as a multivariate Gaussian in the ambient space $\mathbb{R}^D$ and use the corresponding **Mahalanobis distance** as the detection score.

Given K library patches $\{\mathbf{v}_1, \ldots, \mathbf{v}_K\} \subset \mathbb{R}^D$, one estimates:

$$\boldsymbol{\mu} = \frac{1}{K}\sum_{k=1}^{K} \mathbf{v}_k, \qquad \boldsymbol{\Sigma} = \frac{1}{K}\sum_{k=1}^{K} (\mathbf{v}_k - \boldsymbol{\mu})(\mathbf{v}_k - \boldsymbol{\mu})^\top \;\in\; \mathbb{R}^{D\times D}.$$

The Mahalanobis distance of a query feature $\mathbf{f}$ to the class is:

$$d_{\mathcal{M}}(\mathbf{f},\, \boldsymbol{\mu}) = \sqrt{(\mathbf{f} - \boldsymbol{\mu})^\top\, \boldsymbol{\Sigma}^{-1}\, (\mathbf{f} - \boldsymbol{\mu})}.$$

The detection criterion becomes $d_{\mathcal{M}}(\mathbf{f}, \boldsymbol{\mu}) < \tau$. The key is the eigendecomposition $\boldsymbol{\Sigma} = \mathbf{U}\boldsymbol{\Lambda}\mathbf{U}^\top$, which gives $\boldsymbol{\Sigma}^{-1} = \mathbf{U}\boldsymbol{\Lambda}^{-1}\mathbf{U}^\top$. An eigenvector $\mathbf{u}_i$ of $\boldsymbol{\Sigma}$ with a **large eigenvalue** λ_i represents a direction of high within-class variance; $\boldsymbol{\Sigma}^{-1}$ assigns this direction a small weight λ_i^{-1}, making the distance *tolerant* of deviations in that direction. Conversely, an eigenvector with a **small eigenvalue** λ_j (a tight inter-class boundary) receives a large weight λ_j^{-1}, and deviations are *strongly penalised*. Cosine similarity, by contrast, implicitly uses $\boldsymbol{\Sigma}^{-1} = \mathbf{I}$—it weights every dimension identically, as if the class were perfectly isotropic.

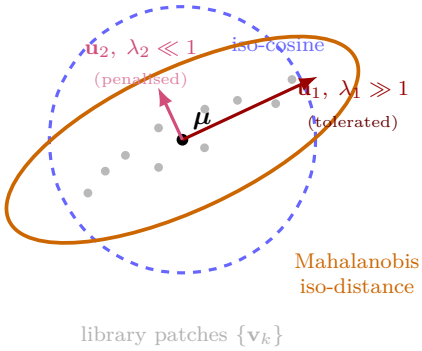


Figure 26: Mahalanobis iso-distance contour (orange ellipse) versus isotropic cosine iso-score contour (blue dashed circle) around the class mean $\boldsymbol{\mu}$. The Mahalanobis ellipse is elongated along the principal eigenvector $\mathbf{u}_1$ (large λ_1, within-class variation, tolerated) and compressed along $\mathbf{u}_2$ (small λ_2, inter-class boundary, penalised). Grey dots are library patch embeddings.

But to better exploit the fact that it is mostly the direction of the vectors that have a meaning we can do the following:

6.3 Fisher-Bingham Distribution on the Hypersphere

The **Fisher-Bingham (FB) distribution** [3, 4] is the natural exponential family on $\mathcal{S}^{D-1}$. Its general form is:

$$p(\mathbf{x}\,;\,\mathbf{A},\mathbf{b}) \;\propto\; \exp\!\big(\mathbf{x}^\top \mathbf{A}\,\mathbf{x} + \mathbf{b}^\top \mathbf{x}\big), \qquad \mathbf{x} \in \mathcal{S}^{D-1},$$

where $\mathbf{b} \in \mathbb{R}^D$ governs the mean direction (first-order term) and $\mathbf{A} \in \mathbb{R}^{D\times D}$ (symmetric) governs anisotropic concentration (second-order term). Setting $\mathbf{A} = \mathbf{0}$ recovers the von Mises-Fisher distribution, confirming that vMF—and by extension the cosine similarity threshold—is the *isotropic special case* of the Fisher-Bingham family.

6.4 Geometric Interpretation: Preferred and Penalised Directions

Both the Mahalanobis distance and the Fisher-Bingham distribution arrive at the same fundamental insight, via different routes: the D dimensions of the embedding space are *not* equally informative for a given class.

- **Preferred (tolerated) directions**: eigenvectors with large eigenvalue of $\boldsymbol{\Sigma}$, equivalently small eigenvalue of $\mathbf{A}$. The class extends far along these axes due to legitimate within-class variation. The detection boundary is lenient: a query may deviate substantially from $\boldsymbol{\gamma}_1$ in these directions and still be accepted.
- **Penalised directions**: eigenvectors with small eigenvalue of $\boldsymbol{\Sigma}$, equivalently large eigenvalue of $\mathbf{A}$. These axes point toward the boundary with other classes. Even a small angular displacement triggers rejection.

This directional structure is not arbitrary—it is a direct consequence of how DINOv3's pretraining has organised the embedding space. Fine-grained within-class variation (changes in texture scale, illumination colour, viewpoint rotation) produces small angular displacements concentrated along a few specific axes. Coarse semantic transitions (feather → bark → water) produce large displacements along entirely different axes. The scatter matrix $\mathbf{S}$ computed from the library patches automatically discovers which axes belong to which category, without any additional annotation.

Cosine similarity with a single scalar threshold τ cannot distinguish these two types of direction. It assigns the same penalty to a deviation of angle δ regardless of whether that

deviation is along the within-class manifold (should be tolerated) or toward a neighbouring class (should be rejected). Mahalanobis and Fisher-Bingham both correct this by individually calibrating the sensitivity of the detector along each eigendirection.

6.5 Practical Considerations

Minimum library size. Reliable estimation of the $D \times D$ covariance matrix $\mathbf{\Sigma}$ requires $K \gg D$. For ViT-L embeddings ($D = 1024$), this is infeasible in the few-shot regime. A standard remedy is **shrinkage regularisation**:

$$\mathbf{\Sigma}_{\text{reg}} = (1 - \alpha)\,\mathbf{\Sigma} + \alpha\,\mathbf{I}, \qquad \alpha \in [0, 1],$$

which interpolates continuously between the full anisotropic model ($\alpha = 0$) and isotropic vMF ($\alpha = 1$). The shrinkage coefficient α can be chosen by cross-validation or set via the Ledoit-Wolf optimal estimator [5]. At $K = 1$, $\mathbf{\Sigma} = \mathbf{0}$ and regularisation forces $\alpha = 1$, recovering the cosine similarity of the current system.

7 Conclusion

RaptorVision demonstrates that a pretrained DINOv3 vision transformer, used purely as a frozen feature extractor, provides a sufficiently rich and semantically structured embedding space to enable practical few-shot object detection without any supervised training. The detection pipeline reduces to three elementary operations: L2 normalisation, matrix multiplication, and thresholded max-pooling—all of which are efficiently executed on GPU hardware. The system is immediately applicable to any visual domain for which DINOv3 embeddings are discriminative, with wildlife camera trap analysis as the primary motivating use case.

The empirical evaluation of Section 5 validates this approach across four ecologically diverse camera-trap datasets. Using only 100 training images and DINOv3-Base at 720-pixel resolution, RaptorVision achieves AUC values between 0.96 and 0.99 across all benchmarks, with optimal decision thresholds consistently in the [0.60, 0.70] range. Cross-dataset transfer experiments further confirm that patch libraries generalise across comparable biomes without retraining.

The analysis of Section 6 further reveals that the cosine similarity criterion is the isotropic special case of a richer family of spherical distributions. Modelling the true anisotropic structure of visual classes via Mahalanobis distance or Fisher-Bingham distributions offers a principled path toward improved detection quality as the patch library grows, without requiring any labelled training data.

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